

SSN COLLEGE OF ENGINEERING (Autonomous)
(Affiliated to Anna University, Chennai)
DEPARTMENT OF CSE

UCS 2511 NETWORK LABORATORY

A10: Flow Control

Faculty In-charge: Dr.N.Sujaudeen, Assistant Professor / CSE

Batch: 2020-2024 Year: III Sem: V Sec: A

Learning Outcome:

To be proficient in developing an application to simulate flow control technique using socket programming in C

Write the code using socket programming in C

1. Establish the basic connection between client and the server.
2. Develop an application to simulate flow control technique using HDLC protocol.
3. Client gets input data from user and divides them as 4 bits into a packet.
4. Add header with sequence number, source and destination address.
5. Send them to server. Before sending each packet ask whether to send or not (to create loss).
6. Server on receiving the packet and by reading the sequence number, acknowledges the client as per Go Back N Protocol.
7. If any packet is missing, based on the acknowledgement received from server, client retransmits it.
8. On Client side, before beginning new transmission, asks whether to close connection.

Example:

Packet2: F-B-1-0101-FCS-F Send or Not: Y Sent Packet2 Packet3: F-B-2-1101-FCS-F Send or Not: N Packet4: F-B-3-1011-FCS-F Send or Not: Y Sent Packet 4 Retransmitting: Packet3: F-B-2-1101-FCS-F Sent Packet 3 Packet4: F-B-3-1011-FCS-F Sent Packet 4	Received: F-B-0-1111-FCS-F Received: F-B-1-0101-FCS-F Received: F-B-3-1011-FCS-F Sending Acknowledgment: ACK2 Received: F-B-2-1101-FCS-F Received: F-B-3-1011-FCS-F Sending Acknowledgment: ACK0
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